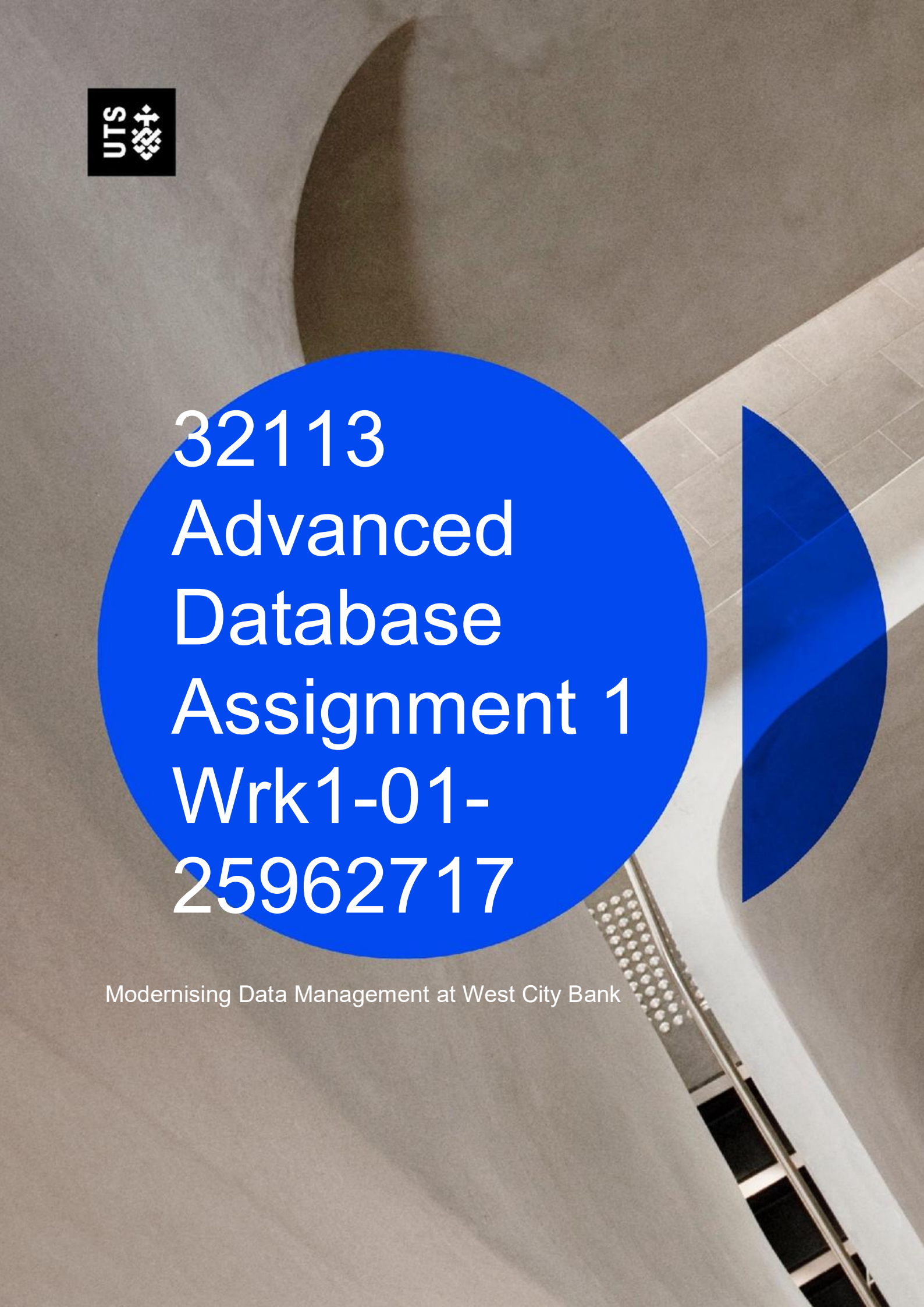


32113 Advanced Database Assignment 1 Wrk1-01-25962717

Modernising Data Management at West City Bank

25659665 Ruotong Yao
25402140 Yi Zhang
25962717 Xiao Li
25112436 Shengxuan Sun
25935901 Viet Hoang Pham
25680909 Dinh Hung Nguyen

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Abstract

Modern banks operate under complex enterprise information environments where a single customer transaction may need to be processed and shared by multiple systems. This report investigates the data management challenges associated with a client opening a new transaction account through a mobile banking application. The traditional integration approaches, including batch-based file transfers and highly coupled point-to-point application programming interfaces (APIs), might lead to data delay, system dependencies, processing bottlenecks and data inconsistency across different systems. These issues can not only impact the reliability of banking operation, but also negatively affect customer experiences.

By using a critical literature review and comparative evaluation, the report evaluates multiple data models and database technologies that can support more scalable and reliable enterprise data integration. In particular, the report compares relational/Third Normal Form (3NF) models, dimensional models or star schema models, Data Vault models, and graph data models, and analyse them in terms of data consistency, integration flexibility, scalability, query performance, and applicability to different banking workloads. Furthermore, the report also explores current data management technologies and architecture methods, including data lakes, data Lakehouse, data fabric and data mesh. Through these comparisons, the report finds that there is no single model or technology can perfectly fit all workloads of enterprise. Relational/3NF is most suitable for core operation transaction; data vault is suitable for historical data integration and lineage; graph model is very effective on fraud detection and anti-money laundering; star schema, on the other hand, demonstrates strong capability for reporting and business intelligence. At the platform level, compared to tradition data lake, Lakehouse provides stronger governance, auditability, and analytical capabilities. Hence, we recommend that the West City should establish a reliable event-driven integration layer, adopt layered data modelling architecture, deploy a Lakehouse-centred analytical platform and gradually introduce governed data access and domain-owned data products. By combining these measures, West City can effectively reduce integration delays and enhance cross-system consistency.

1. Introduction

The banking industry has been more and more dependent on digital platforms and interconnected information systems to provide daily financial services. Activities such as opening new transaction accounts through mobile banking applications may seem very simple from the customers' perspective, however, it may involve several enterprise systems co-operating behind the scenes. In order to meet customers' expectations of quickly completing banking businesses through mobile and online channels, the bank must simultaneously maintain accurate record, manage operation risk, support data analysis and satisfy regulatory obligations (Australian Transaction Reports and Analysis Centre 2026) But the information required by these customer services is often isolated and distributed across multiple applications, departments and databases. Therefore, how to manage data consistently across multiple organizational systems has been a vital concern for enterprise data management.

One of the key challenges in this environment is that information created during a digital banking activity process may be afterwards required by operational processing, customer management, risk assessment, reporting and analytical purposes. These activities impose different requirements on the freshness, consistency, structure and availability of data. Therefore, enterprise integration not only involves data transmission between systems, but also includes ensuring the reliability and availability of information when transferring it among different departments within the organization. Delays, repetitions or inconsistencies may lead to a decline in trust in the organization's data, make downstream processing more complex and affect the response speed of digital services.

West City, a bank located in Western Australia, is currently facing challenges in integrating and synchronizing the data generated during the mobile transaction account opening process. When customers open new accounts through the mobile application, the generated customer and account data must be shared across the bank's multiple inner information systems. If this data transmission process is delayed or fails, it may result in inconsistent records, processing bottlenecks and service disruptions. These issues can negatively affect internal operations and customer experience, especially when the system relies on timely and reliable access to the same information. Currently, West City Bank relies on a hybrid approach of scheduled batch file transmission and point-to-point synchronous APIs. Under this arrangement, disruptions in the downstream system may cause a blocked opening account process. Asynchronous transmission may be delayed, duplicated, processed out of order, or can not immediately detect failure. Event-driven systems reduce coupling by allowing components to communicate asynchronously, but this also brings about complex issues of cross-service consistency and integrity (Laigner et al. 2021). Therefore, the challenge for West City is to improve the management and sharing methods of account opening data while supporting the bank's different operational and analytical needs.

To address this problem is not merely about choosing a more efficient integration mechanism. The structure, storage, processing and sharing methods of enterprise data will significantly affect the scalability, flexibility, consistency and analytical performance of the entire bank data environment. Different data models are designed for different purposes, so there are different trade-offs.

Therefore, this report will critically review four data models: relational modelling in third normal form, dimensional modelling using a star schema, Data Vault modelling and graph modelling. The report will also review four enterprise data-management

technologies: data lakes, data Lakehouse, data fabric and data mesh. These methods will be evaluated by the field of data integrity, synchronization delay, scalability, applicability, flexibility, and implementation complexity, in order to propose appropriate solutions for West City Bank and enhance the overall data consistency and system resilience of its banking enterprise information environment.

2. Business Problem

2.1 Industry Vertical: Banking and Financial Services

Banking and financial services is one of the most data-intensive and heavily regulated sectors anywhere in the world, and Australia is no exception. Retail and business banks handle enormous volumes of sensitive data, customer identity, transaction history, credit risk, product holdings, all of which has to be captured, stored, and reported accurately, with a clear audit trail and within tight timeframes. The sector also carries a lot of legacy baggage: many established banks are still running core banking platforms built in the 1980s and 90s on mainframes or early relational systems, and these now have to somehow coexist with modern digital channels like mobile apps, open banking APIs, and real-time payments.

Australia's regulatory environment only raises the stakes. APRA sets demanding standards for data governance and risk management, while AUSTRAC imposes strict transaction-monitoring and reporting obligations under anti-money-laundering and counter-terrorism financing law. On top of that, the Consumer Data Right requires banks to share customer and product data with accredited third parties on request, in near real time. Put together, these pressures mean data integration, consistency, and timeliness are not just nice-to-haves for a bank, they are compliance-critical. If a bank cannot reconcile its ledger, resolve a customer's identity across its systems, or raise an accurate AML alert within the regulatory window, the consequences are direct: financial penalties, and potentially licensing issues.

2.2 Organisation: West City Bank

West City is a fictitious tier-two Australian retail and business bank serving around 1.8 million personal customers and 140,000 business customers across transaction accounts, savings, home loans, personal loans, credit cards, and business lending. Customers reach West City through four channels: the mobile banking app, internet banking, a network of 90 branches, and a phone-based contact centre.

The bank's core processes span account origination and KYC verification, day-to-day transaction processing, loan origination and servicing, card issuing and fraud monitoring, and regulatory reporting to APRA and AUSTRAC. Each of these processes touches different data: customer master data (name, date of birth, address, tax file number), account and product data (account number, product type, interest rate, balance), transaction data (amount, timestamp, merchant category, channel), and risk data (credit score, fraud flags, AML risk rating).

2.3 Business Problem: System Integration

At the heart of West City's system integration architecture is a real business problem: critical business events aren't propagating reliably across a fragmented technology

estate. The clearest way to see this is through a representative example — a customer onboarding through the mobile app.

When someone opens a new transaction account on West City's mobile app, that single event needs to reach at least six downstream systems, reliably and consistently: the core ledger (to establish the account of record), the fraud and machine learning engine (to risk-score the account before it's activated), the CRM platform (say, Salesforce, to support service and sales workflows), the enterprise data warehouse (for APRA and AUSTRAC reporting), the rewards and loyalty platform (to enrol the customer in relevant offers), and the statement-generation service (to kick off periodic correspondence). Each of these systems keeps its own store of data and updates on its own schedule, and each one depends on getting an accurate, timely picture of what just happened.

The core problem is that West City's current architecture simply cannot guarantee this propagation happens fast enough or consistently enough. Like many incumbent banks, West City has historically relied on one of two approaches: nightly batch file transfers over SFTP, or a patchwork of point-to-point synchronous API calls built up incrementally as new systems came online over the years. Both come with their own failure modes, and the two compound each other. Under the batch model, a new account opened via the mobile app isn't visible to the fraud engine until the next day's batch cycle runs, leaving a period in which newly opened account data may not be available to the fraud and AML monitoring systems. Although AUSTRAC does not prescribe a universal real-time detection threshold, this delay reduces West City’s ability to conduct effective ongoing customer monitoring and should therefore be treated as an operational and compliance risk. Under the point-to-point model, the account-opening process becomes only as reliable as its weakest dependency: an outage in the CRM, for instance, can block account opening entirely, even though the CRM update was never actually required before the account exists on the ledger.

These weaknesses show up as three interconnected kinds of business harm. The first is latency-driven customer experience degradation, chaining together synchronous API calls across six or more systems adds up response time at every hop, leading to mobile app timeouts, incomplete onboarding, and customers dropping off at exactly the point where the bank most needs to convert them. The second is cross-system data inconsistency: because each integration was built independently over time, with its own transformation logic and retry behaviour, a partial failure during propagation can leave an account sitting in the core ledger but missing from the CRM or loyalty platform, leaving the bank with a fragmented, unreliable picture of its own customer. The third is compliance and risk exposure, delayed propagation to the fraud and AML screening engine directly undermines West City's obligation under AUSTRAC to spot and act on suspicious activity promptly, turning what looks like an architecture problem into a regulatory one.

This report looks at how modern integration patterns, including event-driven architecture, Change Data Capture, asynchronous message brokers, combined with the right choice of data model and storage platform, can address this problem. It also evaluates how different data models (Third Normal Form, dimensional/star schema, Data Vault 2.0, and graph models) and storage architectures (data lake versus data Lakehouse) affect the latency and reliability of the integration process.

2.4 Data Use Cases

Building on the business problem above, the following use cases define what an improved integration architecture needs to deliver for West City:

UC1 - Event-driven account propagation: The system should capture the "account opened" event once, at the point it originates, and propagate it reliably to every downstream system without forcing the mobile app to wait synchronously on each one to finish.
UC2 – Timely fraud and AML data availability: Account-opening, customer and KYC data should become available to the fraud and AML monitoring services within a measurable internal service-level objective. The target should be determined through risk assessment and operational testing rather than being presented as a universal AUSTRAC real-time requirement.
UC3 - Consistent multi-system state: CRM, the loyalty platform, the data warehouse, and the statement-generation service all need to eventually converge on the same picture of the customer and their account, with any temporary inconsistency during propagation being visible, bounded, and fixable.
UC4 - Fault-tolerant, decoupled integration: One system going down or degrading would not be able to block the core account-opening transaction. That means decoupling event production from event consumption at an architectural level.
UC5 - Auditability and lineage for regulatory reporting: Data flowing into the enterprise data warehouse for APRA and AUSTRAC reporting needs full lineage, where it came from, when, and what transformations were applied, to support audits and reconciliation.
UC6 - Comparative evaluation of data model and platform impact on integration performance: As a research use case underpinning this report, West City's data function needs evidence-based insight into how different data models (3NF, star schema, Data Vault 2.0, graph) and storage platforms (data lake versus data Lakehouse) affect the latency, throughput, and consistency of the integration pipeline, informing the target-state recommendation developed later in this report.

3. Review

3.1 Data Models

3.1.1 Relational / Third Normal Form (3NF)

The main idea of Third Normal Form (3NF) is to split a large table into several smaller related tables. Different types of data are stored separately, and the tables are connected by IDs or keys. This helps reduce repeated information and keeps the database more organised.

In West City, customer information can be stored in a Customer table, account information in an Account table, and product information in a Product table. When a customer opens a new account, the Account table only needs to store the CustomerID and ProductID. So, the same customer and product details do not need to be stored many times.

The main benefit of 3NF is that it can reduce repeated data and improve data consistency. (Sug 2020) explains that repeated data may cause problems when one copy is updated but another copy is not. Normalisation can help reduce this type of inconsistency.

A simple case is a customer address. If the same address is stored in several account records, the bank may need to update it many times when the customer moves. If one record is missed, different records may show different addresses. With 3NF, the address can mainly stay in the Customer table, so the bank usually only needs to update it once. This makes the data easier to control and reduces the chance of errors.

However, splitting data into many tables can also make some queries more difficult. When the bank wants to see a complete customer record, it may need to join the Customer, Account, Product and other tables together. The more tables involved, the more complex the query can become.

3NF is also not perfect. (Ling, Tompa & Kameda 1981) found that some traditional 3NF designs may still contain unnecessary attributes. So, reaching 3NF does not always mean that the database design has no problems, and careful design is still important.

For West City, 3NF is most suitable for the core banking system and account data, where accurate and consistent information is important. However, 3NF mainly controls how data is stored inside a database. It does not directly solve the problem of moving new account data between the CRM, fraud system, data warehouse and other systems. Therefore, West City still needs other integration technologies to move and synchronise data across different systems.

3.1.2 Dimensional / Star Schema

The main feature of a Star Schema is that the data is organised like a star. There is one main Fact Table in the centre, with several Dimension Tables around it. The Fact Table stores business events or numerical measures, while the Dimension Tables describe the data from different views, such as customer, product, date and channel.

Sanprasit et al. (2021) explain that a typical Star Schema has a fact table in the centre and dimension tables around it. This structure is mainly used for data analysis and OLAP, allowing users to study the same business data from different views.

For West City, an Account Opening Fact Table could record each new account and information such as processing time. It could then connect to Customer, Product, Date and Channel dimensions. This would allow the bank to check how many accounts were opened through the mobile app, which banking products were most popular, or which channels customers used most often.

The main advantage of Star Schema is that it makes reporting and data analysis easier. Users can start from the Fact Table and choose the dimensions they want to analyse. So, managers can compare account openings by month, product, customer type or channel without directly searching through many operational tables. This makes the model useful for management dashboards, business reports and regulatory reporting.

However, Star Schema is mainly designed for analysis rather than daily banking transactions. When a customer opens an account through the mobile app, the bank needs to create and update the account immediately. Star Schema is not mainly built for this type of real-time work. It is more useful after the data has already been collected and prepared for analysis.

Another issue is that designing a Star Schema can take time. The bank needs to decide what should be placed in the Fact Table, which Dimension Tables are needed, and which measures should be analysed. Sanprasit et al. (2021) note that data warehouse and Star Schema design may require expert knowledge and can be time-consuming.

For West City, Star Schema is most suitable for reporting and analysis systems. It can help the bank understand account-opening trends, customer behaviour and business performance. However, it cannot directly solve real-time transaction or system integration problems, so it should work together with other data models and integration technologies.

3.1.3 Data Vault 2.0

Data Vault 2.0 is used to collect data from different source systems and retain historical versions of the data as well as their sources. It consists of three basic structures: Hub, Link, and Satellite. The Hub stores relatively stable business keys such as customer numbers and account numbers; the Link records the relationships between business entities such as customers, accounts, and products; and the Satellite stores the detailed attributes of these entities and their change records. Since business keys, entity relationships, and descriptive information are stored in different tables, when accessing new data sources or adding fields, usually only the corresponding parts need to be expanded, and there is no need to redesign the entire model (Linstedt & Olschimke 2016). This feature makes Data Vault more suitable for large organizations with multiple data sources, high requirements for historical records, and frequently adjusted source systems (Gluchowski 2021).

West City can establish central tables for customers, accounts and products respectively, and then record the account affiliation through the customer-account link table. Contents such as KYC information, account status, risk rating, data source and

loading time can be placed in corresponding satellite tables. Taking mobile banking account opening as an example, after the account opening data enters the Raw Vault, the existing records will still be retained. If the customer updates their address in the future, or if the risk rating and account status change, the system will add a new record instead of overwriting the original content. Thus, West City can restore the state of customer information at a certain point in time and compare the records in the core accounting system, CRM and fraud detection system to find the differences. Research has found that Data Vault can improve schema flexibility when data from changing and heterogeneous sources must be integrated (Nogueira, Romdhane & Darmont 2018). For West City, this capability would primarily support the investigation and reconciliation requirements of UC3 and the auditability and lineage requirements of UC5. However, Data Vault only preserves data that has already been delivered to it; it does not itself propagate the original account-opening event.

Its main advantage for West City is therefore not faster customer-facing transactions, but more reliable historical integration. Source details and load timestamps would support reconciliation, data lineage and regulatory audits. New systems could also be connected through additional links or satellites without requiring a complete redesign of the enterprise model.

However, this flexibility creates additional complexity. A simple customer view may require several hubs, links and satellites to be joined. Data Vault is therefore less suitable for direct reporting and usually needs point-in-time tables, views or downstream dimensional marts. The implementation cost of Data Vault also needs to be taken into consideration. It will generate a large number of data tables. If there is a lack of comprehensive metadata management and automated tools, the subsequent maintenance will become complicated and requires personnel familiar with this architecture to participate. It mainly addresses issues such as data integration, historical retention, and source tracking, and does not guarantee that the opening event will definitely be delivered to all downstream systems. Problems such as message sending failure, incorrect processing sequence, or duplicate consumption still need to be handled by the integration architecture. Based on this, West City can use Data Vault as a historical data integration and audit layer, but it should not replace the core accounting system or event distribution mechanism.

3.1.4 Graph Data Model

The graph data model represents entities such as customers, accounts, or devices as nodes, and the connections or edges between these entities are used to indicate their relationships. Both nodes and relationships can also contain attributes to provide additional information. The relationship model usually requires connecting multiple data tables to identify the relationships between entities, while the graph model directly stores these relationships. Therefore, when the analysis focus is on the paths, networks, or indirect connections between data, the graph model typically has an advantage (Angles & Gutierrez 2008).

In the business scenario of West City, customers, accounts, devices, addresses, transactions and merchants can all be regarded as nodes. The relationships between these nodes can be used to represent that a certain customer "owns" a certain account, a certain account "opens" through a certain device, or two customers "share" the same address. The transfer relationships between different accounts can also be directly recorded in the graph model. Through these connections, the fraud investigation team can more easily discover abnormal situations such as multiple customers using the same device, multiple accounts being interrelated through a series of transactions, or funds circulating between different accounts. If the traditional

relational model is used to analyse the multi-layered associations among customers, accounts, devices and transactions, queries often require continuous connections to multiple tables. As the tracking levels increase, the queries become increasingly complex. Patil et al. (2024) studied the combined application of graph databases and machine learning in bank fraud detection, demonstrating that this method can help identify abnormal connections in the transaction network.

The graph model also facilitates expansion in response to business changes. When new fraud methods emerge or the bank introduces new services, new nodes and relationship types can be directly added. For instance, West City can incorporate IP addresses, phone numbers, and external payees into the analysis in the future, without having to rebuild the entire data model. Compared to the fixed-structure relationship model, this approach is more suitable for handling constantly changing associated data and can be applied to fraud detection, anti-money laundering investigations, entity identification, and customer 360-degree analysis.

However, graph modelling is not equally suitable for every banking requirement. Account balances and financial transactions require tightly controlled updates and clear transactional rules, while regulatory reporting usually relies on predictable aggregation. Relational and dimensional models are more appropriate for these workloads. A graph database also cannot automatically determine whether two similar customer records refer to the same person. Data cleansing, master data management and entity-matching rules are still required before reliable relationships can be created.

Historical tracking is another limitation. Unless timestamps, effective dates or versioned relationships are explicitly included, a graph may show the current network without clearly explaining how it changed. Large distributed graphs may also create partitioning and traversal challenges (Besta et al. 2023). West City should therefore use the graph model as a specialised analytical layer supplied with validated customer, account and transaction data, rather than as its primary system of record. The graph model is therefore most directly aligned with UC2 because it can support fraud and AML relationship analysis.

3.2 Database Technologies

3.2.1 Data Lake

A data lake ingests data from heterogeneous sources in native or near-native format, keeps it in detailed form, and defers structure until the point of consumption. Sawadogo and Darmont (2021) identify this deferral, known as schema-on-read or late binding, as the defining commitment of the concept. It inverts the schema-on-write discipline behind the relational and dimensional models reviewed in Section 3.1: a warehouse requires an agreed target schema before data is admitted, whereas a lake admits data first and negotiates meaning later. The approach emerged in response to the volume, velocity, variety and veracity of big data, which traditional management systems were not built to absorb.

A mature lake is zoned rather than undifferentiated. Sawadogo and Darmont (2021) describe a reference architecture with a raw zone holding data as ingested, a process zone for transformation and analysis, an access zone serving consumers, and a governance zone managing metadata, quality, security and lifecycle across the other three. Storage is usually a distributed file system such as HDFS or a cloud object store, sometimes in multistore configurations combining relational and NoSQL engines. Processing sits outside storage and is performed by engines such as MapReduce, Spark or Flink. Access is mediated by query languages suited to each store, or by rewriting layers that let one query span several engines.

Governance is the zone on which the architecture depends and the one most often underdeveloped. Nargesian et al. (2019) treat lake management as an open research agenda, listing dataset discovery, extraction, cleaning, integration, versioning and metadata management as unsolved problems, because the lake lacks the catalogue a warehouse assumes. Querying massive collections of datasets of unknown structure and quality is a different problem from querying a modelled database.
For West City, a lake provides a single landing surface for data emitted during account opening: core ledger extracts, CRM exports, fraud engine outputs, application telemetry and KYC documents can all be accepted without first agreeing a common schema. Retaining raw payloads also preserves a record of what was emitted at origination, which the current point-to-point architecture does not. The lake is a storage and analytics substrate, not an integration mechanism. It does not deliver events, guarantee ordering or decouple producers from consumers, so UC1 and UC4 fall outside its scope (Laigner et al. 2021).
Its strengths are ingestion flexibility, low storage cost from separating storage and compute, and the absence of upfront modelling when onboarding sources. Its main weakness follows from neglected governance. Sawadogo and Darmont (2021) call the result a data swamp, a repository data enters but cannot usefully leave, and note that governance, quality and security remain among the least addressed topics in the field. Two further limits are structural. Schema-on-read spreads interpretation across every consumer, so teams may read the same record differently. Cloud object stores also offer no cross-key consistency, so multi-object updates are not atomic and readers may observe partially applied writes (Armbrust et al. 2020).

3.2.2 Data Lakehouse

The Lakehouse responds to these deficiencies directly. It keeps the low-cost object storage and open file formats of the lake but adds a transactional metadata layer supplying warehouse-style guarantees, including ACID transactions, schema enforcement, versioning and audit logging, without moving data into a proprietary system (Armbrust et al. 2021). The foundational Lakehouse literature is authored by Databricks, so it is read here alongside independent sources.

Delta Lake is the reference implementation. Armbrust et al. (2020) describe an ACID table storage layer in which the Parquet objects making up a table are tracked by a write-ahead transaction log held in the object store itself and compacted into Parquet. All metadata sits in the object store and concurrency is optimistic, so no coordinating server is required and clients access a table directly through object store operations. That decision yields capabilities an unmanaged lake lacks: time travel to point-in-time snapshots and rollback of erroneous updates; UPSERT, DELETE and MERGE for record-level amendment and compliance workflows; streaming ingestion in which small objects are written at low latency and transactionally coalesced later; schema evolution without rewriting historical files; and audit logging derived from the log itself. Per-file minimum and maximum statistics in the log also permit data skipping and faster metadata operations than enumerating objects.

Tables are conventionally organised through the medallion pattern: a bronze layer of raw append-only records, a silver layer of cleansed, deduplicated and joined data, and a gold layer of business-level aggregates (Microsoft 2026, Powers n.d.). Its architectural significance is that bronze becomes the source of truth from which later layers can be rebuilt. Streaming into these layers is typically handled by Spark Structured Streaming, which incrementalises a static relational query rather than requiring a physical operator graph, and can provide end-to-end exactly-once

processing when checkpointing, replayable sources and idempotent or transactional sinks are correctly configured (Armbrust et al. 2018).

For West City, the Lakehouse fits the analytical and regulatory requirements in Section 2.4 better than a bare lake. Account-opening data consumed from an upstream event stream can be written continuously into bronze and promoted within seconds instead of waiting for an overnight batch cycle, and the transaction log records every version of every table with the operation that produced it, allowing the exact data state behind any regulatory submission to be reconstructed. Like the lake, however, it consumes events rather than distributing them, so the propagation and decoupling requirements remain outside its scope.

Its limitations are acknowledged in the primary source and bear on the consistency requirement in Section 2.4. Delta Lake is serializable within a single table only, since each table maintains its own log; Armbrust et al. (2020) leave sharing a log across tables as future work, noting it would raise contention under optimistic concurrency. A pipeline updating a customer table and an account table therefore cannot commit both atomically. Guarantees spanning heterogeneous stores remain an open research problem (Kraft et al. 2023). Three further constraints apply: streaming latency is bounded by object store performance and cannot reach the milliseconds needed for synchronous authorisation decisions; secondary indexes beyond per-file statistics are unsupported; and log-structured tables require sustained compaction and layout optimisation to preserve query performance. Peer-reviewed evidence on lakehouses in retail banking specifically also remains thin, the literature being dominated by vendor publications and adjacent accounts of event-streaming adoption (Confluent n.d.).

3.2.3 Data Fabric

Data fabric is a more manageable data management architecture. This technology connects data from different environments through a logical data layer, rather than storing all the data in a single repository (Blohm et al. 2024). Data fabric utilizes metadata and automation technologies to establish a unified logical data layer among various data sources, cloud platforms, and local systems, thereby enabling all data to be accessed and managed uniformly.

The traditional data architecture leads to a one-day delay in fraud detection. In addition, the peer-to-peer API may be blocked because the CRM system is down. Data Fabric may integrate CDC and event-streaming services to support near-real-time data dissemination, although the achievable latency depends on the source systems and connectors (Blohm et al. 2024).

Data Fabric may use data virtualisation to access some sources without physically copying the data, while other sources may still require CDC or replication. It can reduce data-access delays, but it does not by itself guarantee real-time fraud screening.

On the security side, the logical data layer of the data fabric can make data lineage transparent from data source to regulatory reporting (Kerner 2025). Even if the data sources are scattered, it is still possible to trace the origin and transformation process of each data element.

Data Fabric can support more consistent security policies and data lineage across distributed systems. However, as a unified data access entrance, data fabric will generate new data risks once it is misconfigured or attacked (Blohm et al. 2024). And building a high-quality logical data layer requires customizing connectors with outdated systems. Deployment costs are high.

3.2.4 Data Mesh

Data Mesh represents a more decentralized data management architecture. Blohm et al. (2024) points out that the purpose of Data Fabric is to integrate data that is scattered across different physical locations, while the purpose of Data Mesh is to transfer the ownership of data to each business team. In this data architecture, the data will be packaged into data products, which include metadata, quality standards, rules, etc. For instance, the anti-fraud team will receive data products that contain anti-fraud content. It follows four core principles: domain-oriented data ownership, Data as a product, Self-service data platforms, and federated computational governance.

Data mesh can also handle the issue where point-to-point apis may be blocked when the CRM system goes down. When a user opens an account, the core ledger team will upload the account opening as a data product to the data stream. From the user's point of view, the user has completed the account opening, rather than needing to wait for the CRM system. Fostyak and Demkiv (2024) argue that the CRM system or the anti-fraud team will obtain this data product and then process the data for their own system business. This architecture decouples the core team from the CRM system, avoiding congestion.

The Data Mesh can also effectively alleviate the problem of data inconsistency. Data Mesh emphasizes that "data is the product". The business teams are responsible for the data products they release. But unlike artificial accountability, Federated Computational Governance can use automated execution rules (Tonnarelli et al. 2026). Therefore, the business teams need to define clear data contracts. For example, the CRM team must update the customer master data three minutes after opening the account with complete fields. If the data product is not updated, the data product of the CRM team is judged to be unqualified and the data product is forbidden to be used. This approach makes data-quality failures more visible and clarifies responsibility, but it does not eliminate temporary inconsistency between systems.

However, data Mesh requires the business team to not only have business capabilities, but also have professional data processing capabilities and data operation and maintenance capabilities. This requires the bank to make changes to the whole business team, and it requires a lot of costs. This will be the biggest obstacle for traditional banks to deploy data Mesh (Adonis 2025).

Moreover, banks need to invest in building data platforms that allow all business teams to self-publish and consume data products. Chintakindhi (2025) point out that if there are no automated tools to enforce federated computational governance, the data Mesh is likely to turn into data silos, isolating the data products of each business team and undermining the overall data interoperability of the bank.

Overall, Data Lake and Data Lakehouse are used to address the issue of massive data storage; Data Fabric and data Mesh can alleviate real-time latency and system decoupling issues. For an old banking system, a single technology cannot solve the business risk (Vayyala 2025). A better approach is to use a layered hybrid architecture. For example, use Data Lake or Data Lakehouse to store massive data; Reduce data reading latency through data fabric; At the same time, the data productization mechanism of data Mesh is used to solve the problems of account opening congestion and cross-system data inconsistency caused by CRM downtime. In this way, the business systems and data storage of outdated banks can be optimized.

4. Findings

4.1 Key data challenges

Modernising enterprise data management at an Australian retail and commercial bank like West City means grappling with a real tension between getting things done operationally, delivering intelligence fast enough to matter, and staying firmly within regulatory bounds. Looking closely at the current data models and database technologies, five main challenges stand out.

The first is a mismatch between propagation latency and the need for real-time risk detection. Legacy batch transfers over SFTP can introduce delays of several hours, sometimes overnight, which means downstream systems like the anti-fraud engine can not evaluate newly opened mobile transaction accounts until well after the fact. This delay weakens the bank's ability to conduct effective ongoing customer monitoring and respond promptly to unusual transactions and behaviour, creating an operational and compliance risk (Australian Transaction Reports and Analysis Centre 2026).

The second challenge is around system coupling and how failures cascade. Because the mobile onboarding journey is wired up through direct, synchronous API calls to secondary systems like CRM and rewards platforms, an outage or slowdown in one of these peripheral applications can ripple backward and block or even abort the core account-opening process, even though that peripheral system was never critical to begin with.

Third, there's the problem of state divergence and partial writes across systems. Without unified transactional guarantees spanning these different platforms, transient failures during data propagation often mean a record commits successfully to the core banking ledger but never makes it into the CRM or reporting stores. The result is fragmented, inconsistent customer profiles.

Fourth is the question of auditability, provenance, and regulatory lineage. APRA's prudential standards, combined with AUSTRAC and Consumer Data Right obligations, demand data lineage that's both end-to-end and tamper-evident. Unstructured storage like a bare data lake simply is not built for point-in-time reconstruction, which raises the bank's compliance risk considerably.

Finally, there's a structural mismatch across very different workloads. No single data model or technology can do it all, handling high-throughput ACID transactions (best suited to 3NF relational models), reconciling historical deltas (Data Vault), analysing complex network relationships (graph databases), supporting dimensional aggregation (star schema), and powering enterprise streaming, all at once.

4.2 Overall Data Model Comparison

The analysis results indicate that these four data models are respectively suitable for different aspects of the data management requirements of West City Bank. To figure out the advantages of different technologies, this section will compare them from the perspectives of operational data integrity, historical data integration, analysis performance, relational analysis, scalability and flexibility.

Evaluation area	Relational/3NF	Dimensional/Star	Data Vault 2.0	Graph
Primary purpose	Operational transactions	Reporting and BI	Historical integration	Relationship analysis

Data integration	Moderate	Limited	Strong	Moderate after entity matching
Historical tracking	Limited unless separately designed	Moderate through slowly changing dimensions	Strong	Requires explicit temporal design
Standard analytics	Moderate	Strong	Limited for direct querying	Limited for aggregation
Complex relationships	Becomes difficult across multiple joins	Limited	Captures relationships but remains join-heavy	Strong
Flexibility	Controlled but relatively rigid	Moderate	Strong	Strong
Scalability	Strong for defined OLTP workloads	Strong for analytical reads	Strong for parallel and incremental loading	Depends on graph structure and distribution
Best West City use	Core ledger and account master	Regulatory and management reporting	Historical integration, lineage and reconciliation	Fraud, AML and Customer 360
Main limitation	Difficult to adapt across heterogeneous systems	Not suitable for operational processing or event propagation	Complex joins and not an event-delivery mechanism	Not suitable for all transactional and reporting workloads

Table 1. Data Model Comparison

The comparisons above demonstrate that these four models can complement each other. Relational models are still the best option for core operation transactions, because it has a clear data structure and can ensure the referential integrity and reliable transactional updates. Dimensional model is more suitable for structured, regularly generated analysis and reports, such as account trend analysis, regulatory reporting and management dashboards. Data vault is more suitable for integrating data from multiple systems while keeping complete historical records and data sources. The graph model provides capabilities that relational and dimensional models do not have, and can be used to identify complex associations between customers, accounts, devices, and transactions.

Therefore, rather than choosing only a single data model for the entire organization, West City should adopt a layered data architecture. Relational model can continue to serve as the authoritative data source for customer and operation transactions; account opening events can be further loaded into the data vault to retain the integrated historical records and data lineage. Data relationships related to customers and transactions can be transmitted to the graph database for fraud detection and anti-money laundering analysis. Verified historical data can also be used to build dimensional data marts, thereby supporting regulatory reports and management analysis.

This combination can simultaneously meet the operational and analytical needs of the West City but still cannot completely solve the fundamental problem of data transmission. Data models determine the organization, storage, and query methods of data, but they cannot guarantee that events will be successfully sent to all relevant systems, nor can they ensure that events are processed in the correct order or avoid duplicate processing. These concerns still require technologies and patterns such as Change Data Capture, message brokers, transactional outbox processing, idempotent consumers and reconciliation monitoring. Event-driven systems can reduce dependencies between applications, but consistency must still be actively managed (Lesniak, Laigner & Zhou 2021). The main finding is therefore that West City needs a

coordinated combination of data models supported by a reliable integration architecture, rather than one model presented as a complete solution.

4.3 Overall Technology Comparison

The four database technologies described in Section 3.2 correspond to different levels of the enterprise data architecture. Data Lake, data Lakehouse, data fabric and data mesh are not mutually exclusive competitive solutions, but complementary solutions for risks such as data storage, integration, latency and maintenance. This is a comparison of the capabilities of these technologies in specific data application scenarios.

Evaluation area	Data Lake	Data Lakehouse	Data Fabric	Data Mesh
Primary purpose	Store the original data	Unified analysis, transactional data storage	Logical data integration and real-time access	Data Productization
Data integration	Strong. Accept any format and there is no need to define the Schema in advance.	Strong. Enforce the Schema during writing	Limited. Need to build custom connectors for legacy system	Defined by the domain team and released in the form of a data product
Real-time Capability	Low	Medium	Strong. Provide real-time access across data sources through the logical layer	Medium. Support asynchronous event-driven sharing of data products
Governance & Auditability	Low. Depend on the separately built metadata layer.	Strong. Native support for ACID transactions, time travel, and audit logs	Strong. Providing transparent data lineage and centralized security control	Medium. Rely on federated computational governance and automation tools
Scalability	High. Strong storage scalability	Scalability for both batch and stream processing	Medium. Rely on the quality of the connector and the complexity of metadata management	Highly scalable at the organizational level, but highly dependent on team maturity
Implementation Complexity	Easy to ingest, but easy to become a data swamp	Medium. requires skills in flow engineering and table maintenance, etc.	High. It is difficult to build a virtualization layer on the legacy system.	Very high. Require organizational culture change
Relevance to West City	Can be used as the original database, but it cannot solve the delay problem.	Can be used for APRA/AUSTRAC compliance reports, historical audits and real-time fraud monitoring modelling	Can be used to solve specific delay problems	Can be used to solve the problems of system coupling and consistency

Table 2. Data technology Comparison

These technologies have a layered structure. Data Lake and Data Lakehouse solve the problems of the analysis storage layer, ensuring that data can be continuously cleaned and used for auditing. The Data Fabric solves the problem of the integration layer, allowing the system to access the data in real time without physically moving the data. The data Mesh solves the problems of the operation layer, transforms data into data products of the business team.

For West City Bank, the Data Lakehouse is very helpful in meeting the requirements of "Compliance Audit and Ancestry Tracking (UC5)". This technology provides version control and time travel. Data Fabric can satisfy the near real-time fraud and AML screening (UC2) ", because the technology can realize logic directly connected. The data product concept of the Data Mesh can meet "Fault Tolerance, Decoupled Integration (UC4)", and users can consume data products asynchronously without waiting for the response from the API.

4.4. Limitations and outstanding issues

While the data models and modern platforms explored here go a long way toward solving the major analytical, storage, and architectural bottlenecks, a handful of fundamental engineering, consistency, and organisational issues remain unresolved.

One is the lack of true distributed ACID guarantees across systems. Storage engines like Delta Lake can offer serializable isolation within a single table, but Atomic coordination across heterogeneous systems is technically possible through protocols such as two-phase commit, but it may introduce blocking, latency and availability trade-offs and is often impractical across legacy systems, SaaS platforms and independently operated services. None of the technologies evaluated here can natively guarantee distributed atomicity across different operational endpoints, committing simultaneously to a relational core ledger, a NoSQL store, and a CRM platform, without resorting to complex coordination protocols that end up hurting availability.

There is also a residual layer of eventual consistency and transient divergence to contend with. Decoupling systems through asynchronous message brokers and Change Data Capture removes the synchronous blocking problem, but it does so by trading it for an eventual-consistency model instead. In-flight events, messages arriving out of order, and asynchronous reprocessing can still create brief windows where data is temporarily inconsistent across the systems consuming it.

Graph analytics brings its own dependency: entity resolution and data cleansing. Graph databases are excellent at rapid, multi-hop pattern detection for fraud surveillance, but they can not resolve fuzzy identities or deduplicate records across different channels on their own. The whole graph model depends on upstream, deterministic Master Data Management pipelines and automated entity-matching rules doing their job properly first.

Then there's the organisational complexity of adopting a Data Mesh. Moving to this model means individual business domains such as CRM, Lending, Core Banking, need to take full ownership of their own data pipelines and the data products they publish. For an established tier-two Australian bank, that is a big ask: it demands significant cross-functional data engineering talent, real cultural change, and self-service infrastructure sophisticated enough to stop domains from just becoming silos in a new form.

Finally, there is the integration overhead and technical debt embedded in legacy systems. Laying a unified Data Fabric or modern Lakehouse connector layer over decades-old core banking platforms means building and maintaining high-overhead custom drivers. And no amount of virtualisation can eliminate downtime, schema instability, or network saturation happening at the legacy source layer itself.

4.5 Recommendations

Based on the comparison results, West City should consider implementing the following recommendations:

Recommendation	Key challenge addressed	Recommended action
Establish a reliable event-driven integration layer	Propagation latency (UC1), multi-system inconsistency (UC3), and system coupling and availability (UC4)	Utilize CDC and reliable event streams to capture and transmit changes in account opening data in real time. At the same time, use transaction outbox pattern to ensure the synchronous writing of business data and events, so that account information can be reliably delivered to all downstream systems without blocking the original account opening system.
Adopt a layered data modelling architecture	Structural mismatch across operational and analytical workloads	Maintain the relational model / Third Normal Form (3NF) and use it as the authoritative data model for operational transactions; utilize Data Vault to support historical data integration and data lineage management; employ graph models for fraud detection and anti-money laundering relationship analysis; and use the star schema to support regulatory reports and management reports.
Implement a Lakehouse-centred analytical platform	Near-real-time fraud and AML surveillance (UC2) and regulatory auditability and lineage (UC5)	Store the original account opening data in a controlled staging area, and utilize Lakehouse tables to implement schema constraints, data quality management, version control, time-travel queries, and governed analytical processing. The Lakehouse should support surveillance and reporting rather than inline operational decision-making.
Introduce governed access and domain ownership gradually	Fragmented data access, unclear ownership, legacy integration overhead and Data Mesh organisational complexity	Pilot metadata management, lineage and governed access through selected Data Fabric capabilities. Introduce domain-owned data products gradually in mature business domains rather than implementing a bank-wide Data Mesh immediately.

Table 3. Recommended Actions

5. Conclusion

This report examines the data management, and system integration challenges that West City Bank faces during the process of opening mobile transaction accounts. The current reliance on batch file transfer and point-to-point synchronous APIs may lead to propagation delays, system dependencies, data inconsistencies and increases compliance risks when account information needs to be shared among multiple downstream systems.

The research results indicate that no single data model or database technology can meet all these requirements. Relational/3NF is still applicable for core operational transactions, while data vault support historical integration and lineage, graph modelling can enhance fraud and anti-money laundering relationship analysis and star schema supports reporting and business intelligence. At the platform level, Lakehouse provides stronger governance, auditability and analytical capabilities compared to traditional data lakes and selected data Fabric and data mesh practices can improve data access, ownership and system decoupling.

Therefore, West City Bank should adopt a layered architecture and be supported by a reliable event-driven integration layer, which uses mechanisms such as CDC, durable event streams and idempotent processing. By using this approach, the organization can reduce integration delays, improve consistency and strengthen regulatory traceability. Future implementation should be carried out step by step, starting from the riskiest integration points and then gradually introducing controlled data access and domain-owned data products as the organization and technical capabilities mature.

References

Adonis, P.H.C.·2025,·'Revolutionizing Data Management in the Financial Sector: A Data Mesh Approach to Governance, Compliance, and Scalability: Challenges, Opportunities, and Practical Implications for Financial Institutions', master's thesis,·NOVA Information Management School, Universidade Nova de Lisboa.

Angles, R. & Gutierrez, C. 2008, 'Survey of graph database models', *ACM Computing Surveys*, vol. 40, no. 1, pp. 1-39. <https://doi.org/10.1145/1322432.1322433>.

Armbrust, M., Das, T., Sun, L., Yavuz, B., Zhu, S., Murthy, M., Torres, J., Hovell, H.v., Ionescu, A., Łuszczak, A., Świtakowski, M., Szafrański, M., Li, X., Ueshin, T., Mokhtar, M., Boncz, P., Ghodsi, A., Paranjpye, S., Senster, P., Xin, R. & Zaharia, M. 2020, 'Delta lake: high-performance ACID table storage over cloud object stores', *Proceedings of the VLDB Endowment*, vol. 13, no. 12, pp. 3411-24. <https://doi.org/10.14778/3415478.3415560>.

Armbrust, M., Das, T., Torres, J., Yavuz, B., Zhu, S., Xin, R., Ghodsi, A., Stoica, I. & Zaharia, M. 2018, 'Structured Streaming: A Declarative API for Real-Time Applications in Apache Spark', *Proceedings of the 2018 International Conference on Management of Data*, Association for Computing Machinery, New York, NY, pp. 601-13.

Armbrust, M., Ghodsi, A., Xin, R. & Zaharia, M. 2021, 'Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics', *11th Annual Conference on Innovative Data Systems Research (CIDR '21)*, Online, viewed 1 September 2026, <https://www.cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf>.

Australian Transaction Reports and Analysis Centre 2026, *Changes to AML/CTF obligations: what you need to do*, viewed 1 September 2026, <https://www.austrac.gov.au/news-and-media/article/changes-amlctf-obligations-what-you-need-do>.

Besta, M., Gerstenberger, R., Peter, E., Fischer, M., Podstawski, M., Barthels, C., Alonso, G. & Hoefler, T. 2023, 'Demystifying Graph Databases: Analysis and Taxonomy of Data Organization, System Designs, and Graph Queries', *ACM Computing Surveys*, vol. 56, no. 2, pp. 1-40. <https://doi.org/10.1145/3604932>.

Blohm, I., Wortmann, F., Legner, C. & Köbler, F. 2024, 'Data products, data mesh, and data fabric', *Business & Information Systems Engineering*, vol. 66, no. 5, pp. 643-52. <https://doi.org/10.1007/s12599-024-00876-5>.

Chintakindhi, S.K. 2025, 'Federated AI governance mesh for multi-cloud platforms: Enabling policy-driven compliance and real-time data trust across autonomous systems', *International Scientific Journal of Engineering and Management*, vol. 4, no. 5, pp. 1-9. <https://doi.org/10.55041/isjem03312>.

Confluent n.d., *KeyBank fuels national digital bank initiative with data in motion and Confluent*, viewed 1 September 2026, <https://www.confluent.io/customers/keybank/>.

Fostyak, M. & Demkiv, L. 2024, 'Development of an AI Domain in a Data Mesh Network for Customer Credit Classification Using Transaction Data', *2024 IEEE 19th International Conference on Computer Science and Information Technologies (CSIT)*, IEEE, Piscataway, NJ, pp. 1-5.

Gluchowski, P. 2021, 'Data Vault as a Modeling Concept for the Data Warehouse', in S. Aier, P. Rohner & J. Schelp (eds), *Engineering the Transformation of the Enterprise: A Design Science Research Perspective*, Springer International Publishing, Cham, pp. 277-86.

Kerner, S.M. 2025, 'Understanding how data fabric enhances data security and governance', *Network World*, 16 June, viewed 1 September 2026, <https://www.networkworld.com/article/4007018/understanding-how-data-fabric-enhances-data-security-and-governance.html>.

Kraft, P., Li, Q., Zhou, X., Bailis, P., Stonebraker, M., Zaharia, M. & Yu, X. 2023, 'Epoxy: ACID Transactions across Diverse Data Stores', *Proceedings of the VLDB Endowment*, vol. 16, no. 11, pp. 2742-54. <https://doi.org/10.14778/3611479.3611484>.

Laigner, R., Zhou, Y., Salles, M.A.V., Liu, Y. & Kalinowski, M. 2021, 'Data management in microservices: state of the practice, challenges, and research directions', *Proceedings of the VLDB Endowment*, vol. 14, no. 13, pp. 3348-61. <https://doi.org/10.14778/3484224.3484232>.

Lesniak, A., Laigner, R. & Zhou, Y. 2021, 'Enforcing consistency in microservice architectures through event-based constraints', *Proceedings of the 15th ACM International Conference on Distributed and Event-based Systems*, Association for Computing Machinery, pp. 180-3, <https://doi.org/10.1145/3465480.3467839>.

Ling, T.-W., Tompa, F.W. & Kameda, T. 1981, 'An improved third normal form for relational databases', *ACM Transactions on Database Systems*, vol. 6, no. 2, pp. 329-46. <https://doi.org/10.1145/319566.319583>.

Linstedt, D. & Olschimke, M. 2016, *Building a scalable data warehouse with Data Vault 2.0*, Morgan Kaufmann, Boston.

Microsoft 2026, *Phase 6: Design Delta Lake architecture*, Microsoft Learn, viewed 1 September 2026, <https://learn.microsoft.com/en-us/azure/databricks/lakehouse-architecture/deployment-guide/delta-lake>.

Nargesian, F., Zhu, E., Miller, R.J., Pu, K.Q. & Arocena, P.C. 2019, 'Data lake management: challenges and opportunities', *Proceedings of the VLDB Endowment*, vol. 12, no. 12, pp. 1986-9. <https://doi.org/10.14778/3352063.3352116>.

Nogueira, I.D., Romdhane, M. & Darmont, J. 2018, 'Modeling Data Lake Metadata with a Data Vault', *Proceedings of the 22nd International Database Engineering & Applications Symposium*, Association for Computing Machinery, New York, NY, pp. 253-61, <https://doi.org/10.1145/3216122.3216130>.

Patil, A., Mahajan, S., Menpara, J., Wagle, S., Pareek, P. & Kotecha, K. 2024, 'Enhancing fraud detection in banking by integration of graph databases with machine learning', *MethodsX*, vol. 12, article 102683. <https://doi.org/10.1016/j.mex.2024.102683>.

Powers, M. n.d., *Building the medallion architecture with Delta Lake*, Delta Lake, viewed 1 September 2026, <https://delta.io/blog/delta-lake-medallion-architecture/>.

Sanprasit, N., Jampachaisri, K., Titijaroonroj, T. & Kesorn, K. 2021, 'Intelligent approach to automated star-schema construction using a knowledge base', *Expert Systems with Applications*, vol. 182, article 115226. <https://doi.org/10.1016/j.eswa.2021.115226>.

Sawadogo, P. & Darmont, J. 2021, 'On data lake architectures and metadata management', *Journal of Intelligent Information Systems*, vol. 56, no. 1, pp. 97-120. <https://doi.org/10.1007/s10844-020-00608-7>.

Sug, H. 2020, 'A method for normalization of relation schema based on data to abide by the third normal form', *WSEAS Transactions on Mathematics*, vol. 19, pp. 216-25. <https://doi.org/10.37394/23206.2020.19.20>.

Tonnarelli, M., van Eijk, T., Kumara, I., Di Nucci, D., Tamburri, D.A. & van den Heuvel, W.-J. 2026, 'Architectural Design Decisions for Federated Computational Governance in Data Meshes', *2026 IEEE 23rd International Conference on Software Architecture (ICSA)*, IEEE, Piscataway, NJ, pp. 143-54.

Vayyala, R. 2025, 'Designing Microservices-Based Data Lake Architectures for Unified Analytics in Legacy-Modern Hybrid Banking Ecosystems', *2025 5th International Conference on Electrical, Computer and Energy Technologies (ICECET)*, IEEE, Piscataway, NJ, pp. 1-7.

Appendix A

Student Name	Allocated Task	Report Section	Completion Date/Time
Xiao Li	Responsible for Abstract, Section 1 Introduction; Section 5 Conclusion;	Abstract, Introduction & Conclusion	15th August
Yi Zhang	Responsible for 3.1.1 Relational / Third Normal Form (3NF) and 3.1.2 Dimensional / Star Schema	Section 3.1.1 & 3.1.2	25th August
Ruotong Yao	Responsible for Data Vault 2.0 and the Graph Data Model, including their concepts, strengths, limitations and application to West City. Responsible for comparing all four data models, recommendation in Section 4.5, and formatting and integrating the references.	Section 3.1,3 & 3.1.4, 4.2, 4,5	28th August
Shegnxuan Sun	Responsible for Data technology 3.2.3 Data Fabric and 3.2.4 Data Mesh and Overall Technology Comparison	Section 3.2.3 & 3.2.4 Section 4.3	1st August
Viet Hoang Pham	Responsible for Data technology 3.2.1 and 3.2.2	Section 3.2.1 & 3.2.2	28th August
Dinh Hung Nguyen	Responsible for Business Problem; Key data challenges and Limitations and outstanding issues	Section 2, 4.1 and 4.4	28th August

Table 4. Individual Contribution Table

Xiao Li



Figure A1. Evidence of communication in Teams group chat

Dinh Hung Nguyen

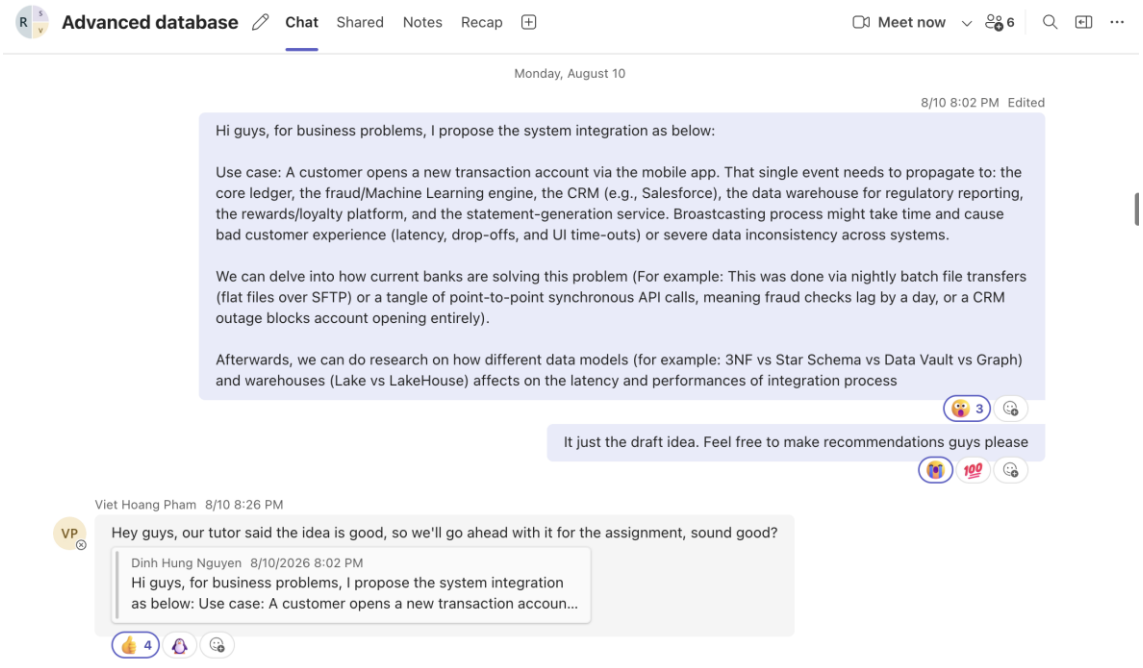


Figure A2. Evidence of communication in Teams group chat

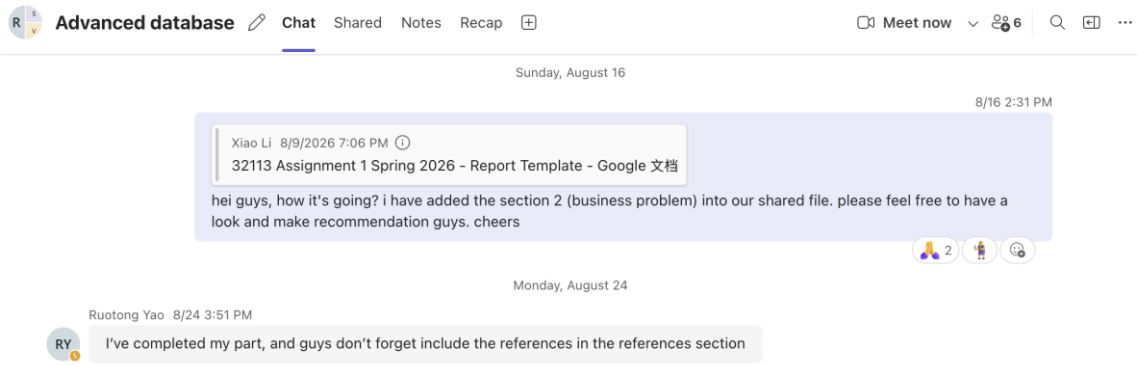


Figure A3. Evidence of communication in Teams group chat

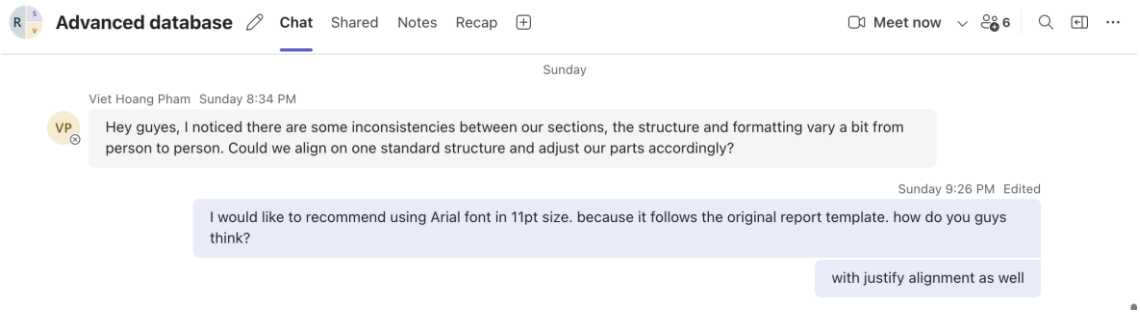


Figure A4. Evidence of communication in Teams group chat

Shengxuan Sun

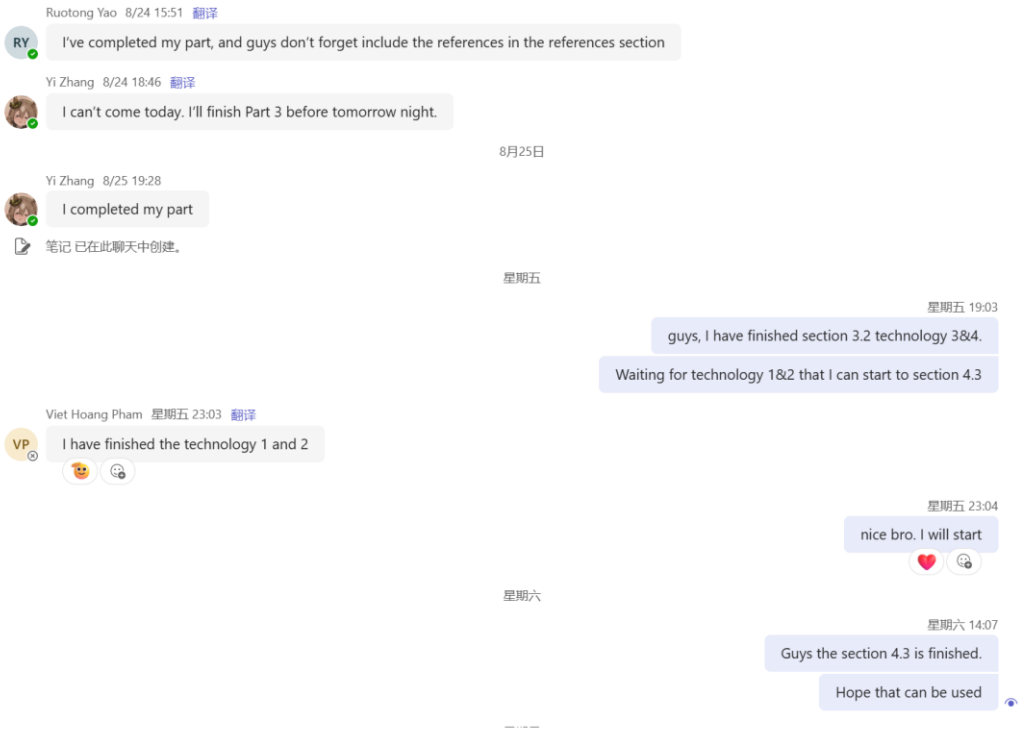


Figure A5. Evidence of communication in Teams group chat

Ruotong Yao

The three attached figures show the Teams group chat, online document revisions and EndNote reference integration.

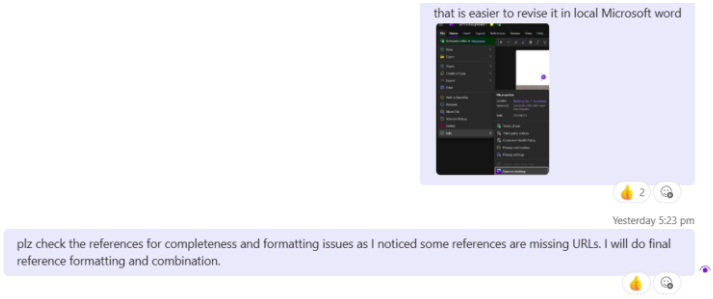


Figure A6. Teams group chat screenshot

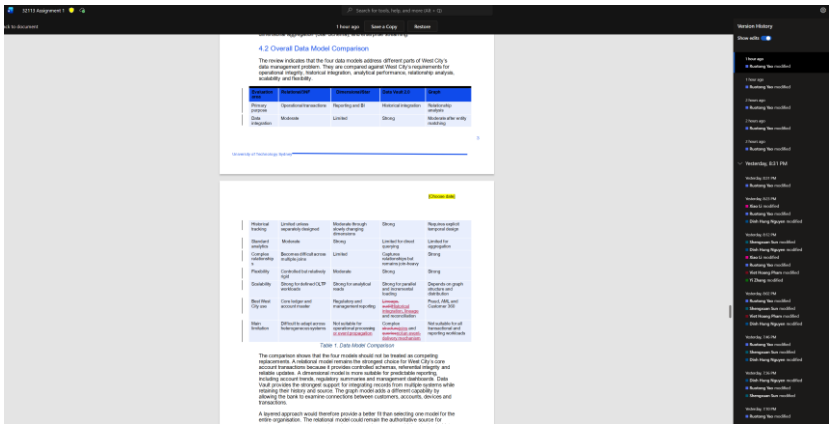


Figure A7. Online document revision history

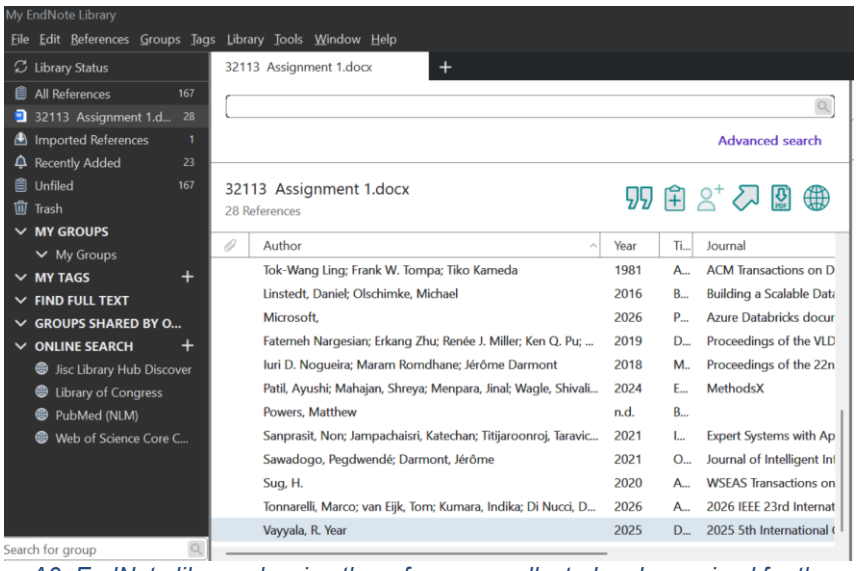


Figure A9. EndNote library showing the references collected and organised for the report

Viet Hoang Pham

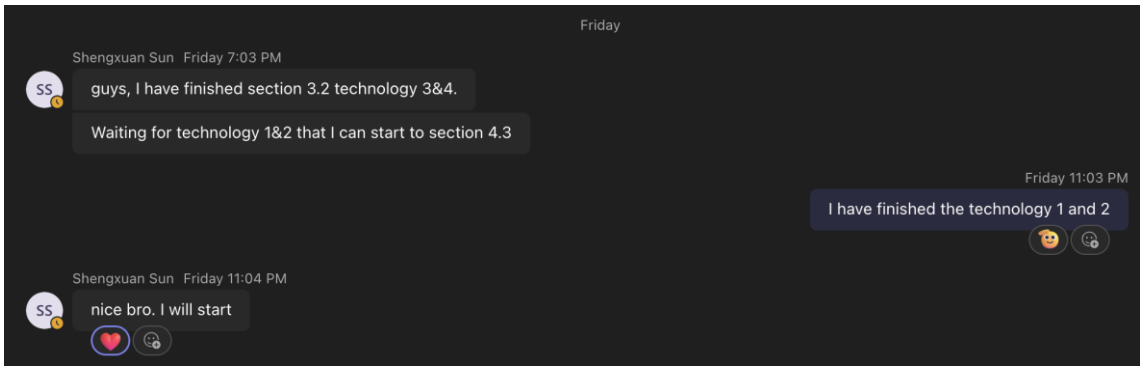


Figure A10. Evidence of communication in Teams group chat

Yi Zhang

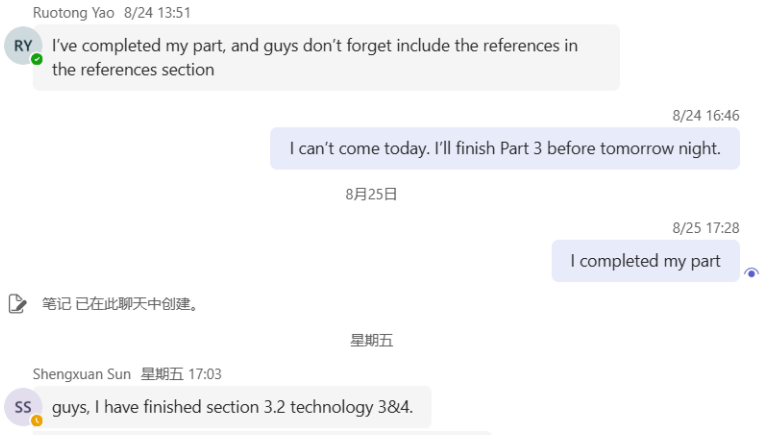


Figure A11. Evidence of communication in Teams group chat

Appendix B

Overall Individual Contribution:

Student ID	Name	Individual Contribution (0-100%)
25402140	Yi Zhang	100
25962717	Xiao Li	100
25659665	Ruotong Yao	100
25112436	Shengxuan Sun	100
25935901	Viet Hoang Pham	100
25680909	Dinh Hung Nguyen	100

Appendix C

Meeting Minutes

GROUP NR.: Wrk1-01-25962717	MEETING NR.: 1	DATE/TIME: 3rd Aug 2026	TEACHER SIGNATURE: B01

ATTENDEES:	ACTION ITEMS:
- Name, ID - Xiao Li, 25962717 - Ruotong Yao, 25659665 - Shengxuan Sun, 25112436 - Yi Zhang, 25402140 - Viet Hoang Pham, 25935901 - Dinh Hung Nguyen, 25680909 **ABSENT:** - …	- Each member will research potential industries and organizations. - Prepare 1–2 possible choices to discuss at the next meeting. - Finalize the selected industry and organization at the next meeting.

MEETING NOTES:

During this meeting, we mainly introduced ourselves, discussed topics that we were interested in, and expressed our opinions on which industry and organization we should choose. We have agreed that we will conduct some research ourselves and then make specific decisions at the next week's meeting.

ASSIGNED RESPONSIBILITIES:	ACCOMPLISHED TASKS:
- All members will research possible industries and organizations before the next meeting - Each member will prepare 1–2 options and briefly explain the reasons for their choices - The group will compare the options and make a final decision at the next meeting	- Introduced all group members - Discussed individual interests and possible project topics - Shared initial ideas about potential industries and organizations

TEACHER FEEDBACK / NEXT STEPS:

Meeting Minutes

GROUP NR.: Wrk1-01-25962717	MEETING NR.: 2	DATE/TIME: 10th Aug 2026	TEACHER SIGNATURE: B01

ATTENDEES:

- Name, ID
- Xiao Li, 25962717
- Ruotong Yao, 25659665
- Shengxuan Sun, 25112436
- Yi Zhang, 25402140
- Viet Hoang Pham, 25935901
- Dinh Hung Nguyen, 25680909

ABSENT:

- …

ACTION ITEMS:

- Each member will complete the assigned part according to the agreed schedule.
- Research and analyse the decided technology
- Link each technology to the needs and operations of the West City.
- Complete the first draft of each specified part within the agreed time frame.
- Review and integrate all parts into the group report.

MEETING NOTES:

In this meeting, we decided to target the banking industry and selected a fictional bank in Western Australia - West City Bank as the target organization. We discussed several possible data technologies and finally agreed to focus on four methods: Data Lake, Data Lakehouse, Data Fabric, and Data Mesh.
The group also discussed the division of responsibilities for every team members. We assigned individual responsibilities and formulated a time schedule. We jointly determined the project deadline and clarified the main tasks that needed to be completed in each stage.

ASSIGNED RESPONSIBILITIES:

- Xiao Li: Abstract, Section 1 Introduction, Section 5 Conclusion
- Ruotong Yao: Section 3.1 Data vault model, Graph database model and section 4 Findings and Recommendations.
- Shengxuan Sun: Section 3.2.3, 3.2.4, 4.3
- Yi Zhang: Section 3.1.1 & 3.1.2
- Viet Hoang Pham: Section 3.2.1 & 3.2.2
- Dinh Hung Nguyen: Section 2 & 4.1 & 4.4
-

ACCOMPLISHED TASKS:

- The banking industry was decided as the target industry.
- West City was selected as the target Organization.
- Data Lake, Data Lakehouse, Data Fabric, and Data Mesh were confirmed as the four technologies
- Individual responsibilities were assigned to the team members
- A time schedule and deadline were established

TEACHER FEEDBACK / NEXT STEPS: